

Enhancer biology and enhanceropathies

Edwin Smith & Ali Shilatifard

Enhancers are *cis*-regulatory elements that enable precise spatiotemporal patterns of gene expression during development and are notable for being able to function at large distances from their target genes. Such regulatory elements often bypass intervening genes and typically comprise binding sites for multiple transcription factors that can also be transcribed by RNA polymerase II (Pol II) to produce noncoding enhancer RNAs (eRNAs). Genome-wide analyses have revealed chromatin signatures of enhancers, such as the enrichment for monomethylation of histone H3 lysine 4 (H3K4me1) and the acetylation or methylation of histone H3 lysine 27 (H3K27). Enhancer signatures have been used to describe the transitions of these regulatory elements from inactive to primed and from activated to decommissioned states during development. New mutations of enhancer sequences and of the protein factors regulating enhancer function in human disease continue to be identified, contributing to a growing class of ‘enhanceropathies’.

The term ‘transcriptional enhancer’ was coined over 30 years ago by Schaffner and colleagues to describe the properties of a tandem 72-bp repeat from SV40 polyomavirus¹. This element could increase the expression of the rabbit β -globin gene in HeLa cells by two orders of magnitude when linked in *cis*¹. Parallel studies with the 72-bp repeat suggested that enhanced transcription could occur in combination with a variety of known promoters². A low level of transcription was observed even when the element was included in a plasmid with no known promoter, indicating that it was able to recruit Pol II². This element could function in either orientation and at variable distances upstream or downstream of the *cis*-linked gene^{1,2}. These properties quickly came to comprise the standard definition of enhancers as studies moved from viruses to metazoan development.

The first eukaryotic enhancer to be described was the E μ enhancer of the immunoglobulin heavy chain (IgH) gene^{3–5}. An immunoglobulin polypeptide chain results from a DNA recombination that joins non-contiguous segments of the variable and constant regions of immunoglobulin. Chromosomal rearrangements that place the variable region near the constant region are required for high levels of expression, even though the promoter-proximal sequence in front of the variable region is unaltered by these rearrangements. In Burkitt’s lymphoma, high *MYC* expression was known to result from chromosomal translocations that place the *MYC* gene in the same proximity to the constant region as occurs during immunoglobulin rearrangements⁶. Since the implicated region near the IgH gene was located downstream of the affected promoter, it was suspected to contain an element with properties similar to those of the SV40 enhancer. Using heterologous promoters, a segment matching all of the abovementioned criteria for enhancers was mapped to an intron between the heavy-chain joining (J_H) and the μ constant regions^{3–5}.

Although the first enhancers were described in mammalian cells, the mechanistic understanding of enhancer biology was greatly aided by pioneering studies using *Drosophila* embryos. The patterning of the *Drosophila* embryo is set up by gradients of maternally deposited factors. These gradients set broad patterns of transcription factors (the gap genes), which set up seven stripes, each about four cells wide, of expression for the pair-rule genes *even-skipped* (*eve*), *fushi tarazu* (*ftz*), *runt* (*run*) and *odd-skipped* (*odd*)^{7,8}. The seven stripes of expression can result from a combination of an individual enhancer that confers partial expression in all seven stripes and additional enhancers that are required for proper expression in specific stripes^{7,8} (Fig. 1). The stripe enhancers can be found close to the promoter or tens of kilobases upstream or downstream (Fig. 1)^{7,9}. The studies in *Drosophila* embryos provided detailed mechanistic insight into how multiple activators and repressors modulate an individual enhancer and how multiple enhancers determine a single gene’s precise pattern of expression during development.

In this Review, we discuss the classic studies elucidating enhancer function and their misregulation in developmental defects, and we place these findings in the context of more recent discoveries made possible by technological developments. These recent studies confirm the importance of enhancers in orchestrating a transcriptional network during development and point to the fact that the misregulation of enhancer properties is central to the pathogenesis of certain diseases, a class of conditions we refer to as enhanceropathies.

Long-distance enhancer-promoter communication

The study of endogenous enhancers in metazoans led to the realization that enhancers often function at very large distances from a promoter. Even in the relatively compact genome of *Drosophila melanogaster*, an enhancer for the *cut* gene (*ct*) was found 81.5 kb upstream from the transcription start site. This enhancer regulates the specific expression of *ct* in a few hundred cells of the *Drosophila* wing imaginal disc, which contains over 50,000 cells¹⁰. These few hundred cells at the dorsal ventral wing margin will form the sensory bristle cells of the adult wings¹⁰. Defects in

Stowers Institute for Medical Research, Kansas City, Missouri, USA.
Correspondence should be addressed to A.S. (ash@stowers.org).

Received 6 January; accepted 6 February; published online 5 March 2014;
doi:10.1038/nsmb.2784

ct expression in the wing margin lead to visible nicks at the edges of the wing. Mutant alleles of *ct* were obtained when the *gypsy* retrotransposon was inserted between the *ct* wing-margin enhancer and the *ct* promoter (Fig. 2). The *gypsy* element is able to act as a blocker of enhancer-promoter communication because of the presence of 12 binding sites for the insulator protein suppressor of Hairless wing [Su(Hw)]¹¹. The *gypsy*-dependent disruption of *ct* expression and a sensitized allele of *su(Hw)* formed the basis of a genetic screen to identify proteins involved in *ct* wing-margin enhancer function (Fig. 2). Two genes identified in this screen, *Chip* and *Nipped-B*, enhanced the nicked wing phenotype and were found to be general regulators of enhancer-promoter communication^{12–15}. *Chip* encodes a protein homologous to mammalian LDB1 (refs. 12,16). *Chip* and LDB1 have a self-interaction domain that facilitates cooperative binding of transcription factors to DNA and helps bridge enhancer-promoter interactions^{17–20} (Fig. 3).

Nipped-B encodes a protein homologous to the mammalian cohesin-loading factor NIPBL^{14,15}. It was proposed that the ability of the cohesin complex to hold together sister chromatids could also be used to form or stabilize loops of chromatin that bring enhancers and promoters in close proximity¹⁴. Cohesin has subsequently been validated as a major regulator of enhancer-promoter interactions in both *Drosophila* and mammalian cells, using RNA interference (RNAi), chromatin immunoprecipitation sequencing (ChIP-seq) and chromatin conformation capture (3C) assays^{21,22}. Therefore, long-range enhancer-promoter communication could result from the combination of *Chip*/LDB1's bridging of transcription factors at enhancers and promoters, and the ring-like cohesin's helping to hold the enhancer and promoter in close proximity (Fig. 3 and Table 1).

Many models have been proposed to explain how enhancers and promoters find each other, including tracking mechanisms starting at the enhancers and moving to the promoters; looping out of intervening DNA such that enhancers and promoters are brought in close proximity; and a combination of looping and tracking^{23,24}. Although looping has been demonstrated by 3C and related assays, tracking mechanisms from the enhancer to the promoter could be used to set up the initial enhancer-promoter interaction^{23,24}.

LCRs, HOT regions and the many classes of enhancers

One of the most widely studied examples of long-range enhancer-promoter communication is the β -globin locus. Over 50,000 babies are born each year with thalassemias, diseases resulting from insufficient hemoglobin function²⁵, among which α - and β -globin thalassemias are the most common. A subset of β -thalassemias was found to result from the deletion of a large region upstream of the β -globin locus that leaves the β -globin coding region intact^{26–28}. DNase I hypersensitivity, already known from the SV40 and IgH enhancer studies to be a feature of enhancers, was used to define this region, which became known as the β -globin locus control region (LCR)^{26–28}. The β -globin LCR contains five hypersensitive sites, one of which behaves as a classical SV40-like enhancer in reporter assays, while two others enhance transcription only when integrated into chromatin²⁸. Chromosomal regions with multiple elements required to confer position-independent and cell type-specific expression

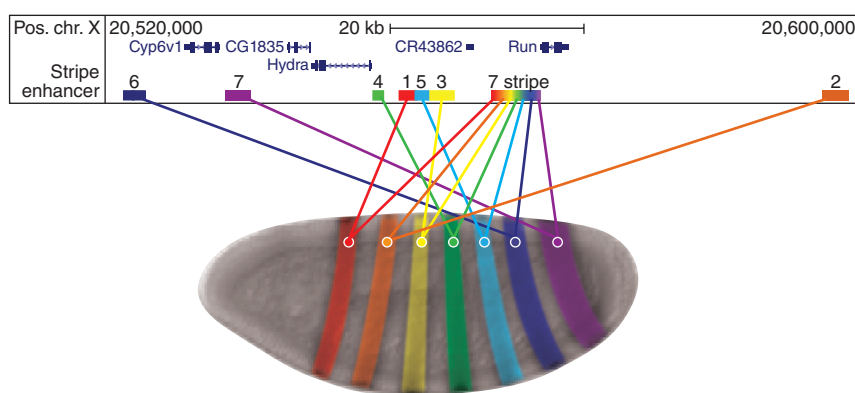


Figure 1 Multiple enhancers regulate a single gene's expression pattern. A schematic of the *Drosophila runt* locus (*run*) and surrounding regions is shown. Exons (vertical bars) and introns (thin lines with arrows indicating direction of transcription) of genes are indicated. A '7-stripe enhancer' located near *run* is able to confer expression of a reporter in all seven of the stripes seen at this stage in early cellular blastoderm embryos. Additional enhancers, which can be found far upstream or downstream of *run*, can each confer expression of a reporter in an individual stripe. Therefore, achieving a full wild-type expression pattern often results from the combined effect of multiple enhancers. Embryo image modified from ref. 7 and reproduced with permission from *Development*.

became defining features of LCRs, which were also referred to as super-enhancers^{29,30}. LCRs were subsequently described for several other loci, including single genes such as the lysozyme gene and the human growth hormone locus, which, like the β -globin locus, contains several genes controlled by one LCR^{31,32}.

Bioinformatic identification of HOT regions. Genome-wide mapping of transcription factors and DNase hypersensitivity have led to novel terms for enhancer-like sequences such as the highly occupied target (HOT) regions of DNA in *Drosophila*. These regions are occupied by eight or more transcription factors and are hypersensitive to DNase I digestion³³. However, only 20% of the identified HOT regions corresponded to previously known enhancers³⁴. When 108 of these HOT regions were used in reporter constructs, the majority were found to confer enhanced expression in specific regions of the embryo³⁴. In the case of *Blimp-1*, a nearby HOT region could account for three of the four stripes of *Blimp-1* expression in the embryo³⁴.

Super-enhancers and stretch enhancers. In a pattern similar to that of the HOT DNA in *Drosophila*, the unusual co-occurrence of clusters of multiple transcription factors, although spread over a larger segment of DNA, was recently reported for a subset of enhancers in mammalian cells^{35,36}. These so-called 'super-enhancers' are most easily distinguished from other enhancers by the very high occupancy of Mediator, a Pol II coactivator that together with cohesin is involved in enhancer-promoter communication^{21,36}. Similar 'transcription initiation platforms', or TIPs, were also identified from bioinformatic analysis as regions that have high transcription-factor occupancy and are associated with active genes³⁷. Both super-enhancers and TIPs can be several kilobases in length. Bioinformatic analysis based largely on histone modifications also led to the identification of extended or 'stretch' enhancer regions³⁸, which are >3 kb wide. Some of the new super-enhancers, TIPs and stretch enhancers defined by ChIP-seq also correspond to characterized LCRs, the original 'super-enhancers' defined by functional studies^{29,30,36,37,39}. However, the vast majority of bioinformatically defined TIPs, stretch enhancers and super-enhancers lack any functional characterization, so it is presently unclear to what extent they constitute novel LCRs.

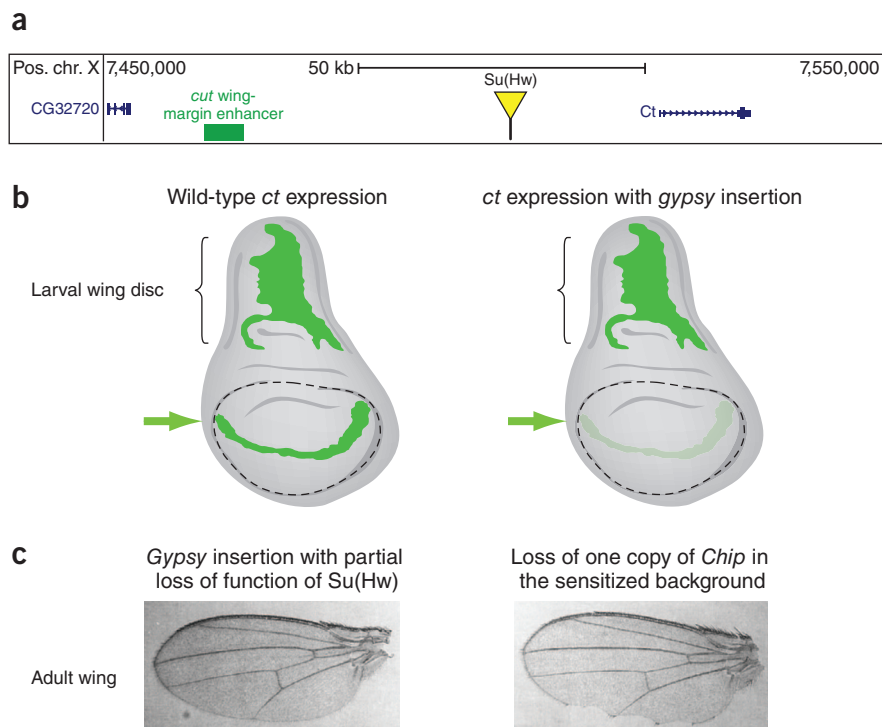


Figure 2 A genetic screen for long-range interactions between enhancers and promoters. **(a)** Schematic of the *cut* (*ct*) locus and the surrounding region. An enhancer conferring expression of *ct* in the wing margin (green rectangle) is located 81.5 kb upstream of *ct*. Insertion of a *gypsy* (yellow triangle) retrovirus anywhere between *ct* and the wing-margin enhancer affects expression of *ct* in the wing margin because of the presence of binding sites within *gypsy* for the insulator-enhancer blocking protein suppressor of hairy wing (Su(Hw)). **(b)** Cartoon of *ct* expression patterns in the larval wing imaginal discs of *Drosophila*. A band of *ct* expression (arrow) across the wing pouch (dashed line) seen in the wild type (left) is absent when *gypsy* is inserted between the wing margin enhancer and *ct* (right). However, another patch of *ct* expression (indicated by a bracket) in the wing disc is not affected by the *gypsy* insertion. Wing disc images modified from ref. 130. **(c)** Left, the sensitized background (*ct^{L-32}; su(Hw)^{e2}*) for a genetic screen for factors affecting enhancer-promoter communication at the *cut* locus (*ct*). Despite the presence of *gypsy*, few nicks are seen in the adult wing owing to the partial loss-of-function allele of the insulator protein Su(Hw) used in the screen. Right, when one copy of *Chip* (*Ch*) is mutated, nicks are more frequent (*ct^{L-32}; Ch^{pe5.5/+}/su(Hw)^{e2}*). Wing images republished, with permission of the Genetics Society of America, from ref. 13.

Shadow enhancers. It is curious that more extended LCRs, TIPs, stretch enhancers and super-enhancers have been described in mammals but not in *Drosophila*. This could reflect a difference in genome organization, with the extended enhancers consisting of a collection of multiple HOT regions within a cluster several kilobases in length. In the relatively compact genome of *Drosophila*, it may be more efficient to spread enhancers around in any available space, whereas in mammalian cells the clustering of *cis*-regulatory elements within an extended enhancer could occur in gene deserts located megabases away from the associated genes. This clustering could help maintain open chromatin, act cooperatively in forming interactions with the promoter to give a more robust expression pattern, or perhaps ‘lock in’ the active state of expression for specific genes. In *Drosophila*, this robust expression might be similarly achieved with shadow enhancers^{40–42}.

The term ‘shadow enhancer’ was used to describe seemingly redundant enhancers identified for genes for which known enhancers existed⁴³. Before the development of genome-wide ChIP data, enhancers were typically identified by trial-and-error cloning of genomic fragments from within and around genes into a reporter construct. Expression of the reporter was compared to the gene’s endogenous expression as assessed by *in situ* hybridization. When a gene’s

expression pattern could be explained (e.g., all seven stripes of *runt* showed strong expression; **Fig. 2**), the set of enhancers for that gene would be considered complete. Shadow enhancers were discovered when genome-wide profiling of transcription factors showed a remarkably similar (‘like a shadow’) combinatorial pattern of transcription factor occupancy at a region distinct from a previously characterized enhancer⁴³. When tested in a reporter assay, this shadow enhancer gave the same pattern as the previously known enhancer⁴³. Careful examination of the timing for achieving a truly endogenous expression pattern in a large number of embryos demonstrated that both the previously characterized enhancer and the shadow enhancer were needed for a timely and reliable expression pattern to occur⁴². The requirement for both enhancers is best seen under suboptimal growth conditions⁴⁴.

Fail-safe or split enhancers. The seemingly polar opposite of the ‘redundant’ shadow enhancers could be the ‘fail-safe’ enhancer elements of the *Aicda* gene. *Aicda* is a cytidine deaminase involved in hypermutation and immunoglobulin class switching. Off-target activities of *Aicda* are proposed to be responsible for the majority of *IGH-MYC* rearrangements in Burkitt’s lymphoma⁴⁵. Minimizing the oncogenic activity of *Aicda* requires tight regulatory control⁴⁵. For example, deletion of either the E1 or E2 *Aicda* enhancer resulted in severe loss of *Aicda* expression⁴⁶, suggesting that *Aicda* levels are kept under a control akin to the ‘two-man rule’ (that acts as a safeguard to prevent an accidental nuclear weapon launch on a submarine). Having the enhancer function split between two or more regions means

that each would have to come together with the promoter to activate the gene. We foresee the use of such fail-safe enhancers in the tissue-specific regulation of the expression of genes whose misexpression could be lethal or oncogenic.

Chromatin signatures in different enhancer states

Genome-wide histone modification ChIP data sets have led to the identification of chromatin signatures for enhancers. The chromatin mark that most reliably predicts enhancers is the relative enrichment of the monomethylated compared to the trimethylated form of H3 lysine 4 methylation (H3K4me1 and H3K4me3, respectively)⁴⁷. Unlike the relatively sharp peak of H3K4me3 at transcription start sites, the H3K4me1 enrichment at enhancers can be quite broad, extending a kilobase or more to either side of the transcription factor-binding sites⁴⁷. The additional presence of H3K27ac at enhancers was found to correlate with a higher gene expression of nearby genes⁴⁸. Using multiple progenitor and differentiated cells, it was shown that the enhancers associated with lineage-specific genes lacked H3K27ac in the progenitor state but gained it during differentiation⁴⁸. Further studies showed that H3K27me3 was found with H3K4me1 at enhancers in embryonic stem cells (ES) and that this combination tended to be associated with

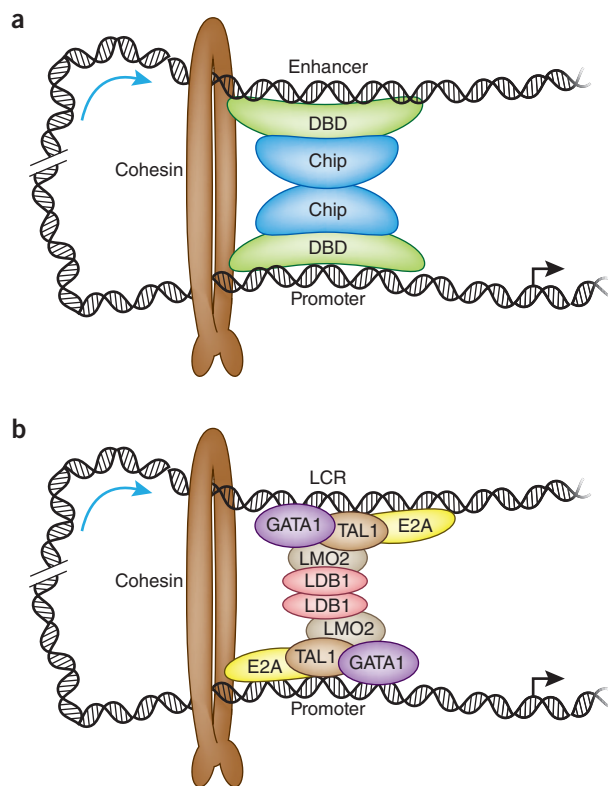


Figure 3 Cohesin and Chip/LDB1 support enhancer-promoter communication in flies and mammals. **(a)** The genetic screen shown in **Figure 2** identified two factors that are generally required for enhancer-promoter communication in flies: Nipped-B (not pictured), the cohesin-loading factor; and Chip, the homolog of the mammalian protein LDB1. Chip is thought to interact with DNA-binding transcription factors (DBDs) at enhancers and promoters and through a self-interaction domain to help bridge enhancer-promoter communication. The ring-like cohesin structure is thought to hold the enhancer and promoter together, a function analogous to its role in sister chromatid cohesion in mitosis. **(b)** Mammalian LDB1 is required for long-range enhancer-promoter communication between the β -globin LCR and its promoter. Shown is a model for the DNA-binding factors GATA1, TAL1 and E2A, the Lim domain protein LMO2, and the self-associating LDB1 at the β -globin locus. Blobel and colleagues tested the proposed role of Chip/Ldb1 in mammalian cells by tethering the self-association domain of LDB1 to a promoter using TALEN zinc fingers, bypassing the requirement for GATA1 to induce β -globin expression¹⁸. Cartoon adapted from ref. 18.

homeotic (Hox) gene regulation^{58,59}; and MLL3 and MLL4 implement H3K4me1 at most enhancers^{51,52}.

Implementing H3K27 acetylation at active enhancers. The H3K27ac mark found at active enhancers is implemented by CBP in *Drosophila* and by its two paralogs CBP and P300 in mammals^{60,61} (**Fig. 4b**). CBP and P300 are major transcriptional coactivators that acetylate a number of substrates in addition to histones⁶². CBP and P300 have many overlapping functions, but they also have nonredundant functions⁶³. CBP was identified on the basis of its physical interaction with cyclic AMP response element-binding protein (CREB), and P300 was identified on the basis of its physical interaction with adenoviral early protein E1A, although both CBP and P300 can interact with either CREB or E1A. Though CBP and P300 are found at active promoters, their occupancy at intergenic or intragenic regions has served as a useful and convenient marker for enhancers in both *Drosophila* and mammalian studies^{47,51,64,65}.

Implementing H3K27 methylation at poised enhancers. The H3K27me3 that marks the poised enhancers of lineage-specific genes in stem cells is implemented by Enhancer of zeste (E(z)) in *Drosophila* and by Ezh1 and Ezh2 in mammals (**Fig. 4a**). E(z) and its homologs function in H3K27 methylation as part of a multisubunit complex called Polycomb repressive complex 2 (PRC2). PRC2 was discovered for its role in homeotic gene repression⁶⁶. The combination of Trx-mediated gene activation and Polycomb-mediated gene repression controls the regulation of *Hox* and many other developmentally regulated genes in metazoans⁶⁷. Among the ten or so subfamilies of the SET domain-containing proteins, the Set1-MLL and E(z) subfamily enzymes are the most similar in sequence⁶⁸. Accordingly, unlike many other SET domain-containing proteins, both the Set1-MLL and E(z) family members require protein complexes for their methyltransferase activity *in vitro* and *in vivo*⁶⁸.

Transitioning from poised to active enhancers. The transition from poised to active enhancers may be aided by the H3K27 demethylase UTX, which is found in the Trx branch of COMPASS in *Drosophila* and the MLL3 and MLL4 branches of COMPASS in mammals^{51–53}. ChIP-seq of Utx in *Drosophila* shows that it is found at putative enhancers along with Trx⁵¹. Depletion of Trx, which leads to the destabilization of Utx, leads to increased H3K27me3 and decreased H3K27ac at enhancer-like chromatin⁵¹. Genome-wide studies in mouse ES cells demonstrate that the H3K27ac enzymes, CBP and P300, occupy both poised and active enhancers^{49,69}. Therefore, the transition to the acetylated state found at active enhancers may involve signaling to CBP and P300 or Ezh2 to change the balance of enzymatic activities to favor the acetyltransferase⁷⁰.

genes involved in cellular differentiation processes⁴⁹. Upon differentiation of ES cells, a subset of these enhancers transitioned to the active chromatin signature⁴⁹. Thus, enhancers were classified into poised and active states on the basis of the status of the H3K27 residue, while both classes maintained enrichment for H3K4me1 (refs. 48,49) (**Fig. 4a,b**). During differentiation, however, many enhancers lose H3K4me1, in a process called enhancer decommissioning⁵⁰ (**Fig. 4c**). This dynamic state of histone modifications on enhancers results from the action of a variety of histone-modifying enzymes (**Table 1**).

Implementing H3K4 monomethylation at enhancers. The monomethylation of H3K4 at enhancers is largely mediated by the Trx-MLL3-MLL4 branch of the COMPASS family^{51,52} (**Fig. 4a,b**). Originally identified in yeast as the first H3K4 methylase, COMPASS is a multisubunit complex containing the histone H3K4 methyltransferase Set1 (ref. 53). In yeast, all histone H3K4 methylation states are implemented by Set1 within COMPASS⁵³. In *Drosophila*, there are three Set1-related enzymes, Set1, Trithorax (Trx) and Trx-related (Trr)⁵³. *Drosophila* Set1 is a true ortholog of Set1 and is found in a complex with a one-to-one subunit correspondence to yeast COMPASS^{54,55}. Trx, discovered through genetics for its role in the regulation of homeotic genes, is found in a complex with a set of core subunits found in all versions of COMPASS as well as some unique subunits such as the tumor suppressor Menin^{55,56}. Trx was identified on the basis of its homology with Trx⁵⁷ and forms a COMPASS-like complex with its own unique subunits, including the H3K27 demethylase Utx⁵⁵. Each of the three COMPASS branches in *Drosophila* is represented by two paralogs in mammals: mammalian Set1A and Set1B are related to *Drosophila* Set1; MLL1 and MLL2 are related to Trx; and MLL3 and MLL4 are related to Trr⁵³. Several functions conserved between the *Drosophila* and mammalian complexes have been demonstrated: the Set1A and Set1B complexes are responsible for bulk H3K4me3; MLL1 and MLL2 have a prominent function in

Table 1 Examples of transcription factors and other proteins acting at enhancers

Functional class	Factor	Proposed enhancer function	Refs.
Pioneer transcription factors	GATA	GATA- and zinc-finger-containing family of DNA-binding proteins. GATA1 is required for erythropoiesis.	108,131
	Forkhead (FOX)	Forkhead box–containing family of proteins. Can bind nucleosomes with a winged-helix motif similar to histones.	132–134
	Pu.1	ETS domain–containing protein. Required for myeloid and B-lymphocyte development.	135,136
	Zelda	Multi-zinc-finger protein that marks enhancers and promoters in <i>Drosophila</i> embryos before zygotic transcription begins.	34,137–139
Chromatin topology	Nipped-B/NIPBL and cohesin	Nipped-B loads the ring-like cohesin to lock in enhancer-promoter communication.	140
	Chip/LDB1	Uses a self-association (SA) domain to help bridge enhancer and promoters by interacting with factors such as GATA1 during erythropoiesis.	19,20,141
	Su(Hw), CTCF	Insulator proteins that help define topological domains. Can have enhancer-blocking activity if located between the enhancer and the promoter.	142–144
Histone modifiers	Trr, MLL3, MLL4	Implement H3K4me1 at most enhancers in the poised and activated states. Function within COMPASS. Frequently mutated in cancer.	51,52,145
	EZH2	Implements H3K27me3 at poised enhancers.	146
	CBP, P300	Implement H3K27ac at active enhancers.	60,146
	LSD1	Removes H3K4me1 to inactivate enhancers	50
	UTX	Subunit of Trr/MLL3–MLL4/COMPASS complex that demethylates H3K27me3 at enhancers. Frequently mutated in cancer.	51
	GCN5, PCAF	Implements H3 acetylation as part of SAGA.	147
	MOF	Implements H4K16ac at enhancers and promoters.	148
Other	Mediator	Large multisubunit complex involved in many stages of transcription. Facilitates enhancer-promoter communication by interacting with enhancer-bound factors and RNA Pol II at the promoter.	21
	ELL3	Transcription elongation factor found at poised and active enhancers in mouse embryonic stem cells. Like Mediator, ELL3 may be able to stabilize Pol II at promoters through its ability to interact with Pol II.	86
	CHD7, BRG1	Nucleosome remodeling factors found at enhancers. May facilitate binding of pioneer factors to enhancers. Required for enhancer-promoter communication.	36,149
	BRD4	Associates with acetylated histones and other proteins. BRD4 occupancy used as marker of active enhancers. Related BRD3 associates with acetylated GATA1 in erythroid cells.	150

Enhancer decommissioning. Genome-wide studies have shown that during differentiation of ES cells many enhancers become silenced, losing the histone modification signatures characteristic of poised and active enhancers⁵⁰ (Fig. 4c). For example, H3K4me1 is removed from the enhancers for *Lefty1* and *Sox2*. The H3K4 demethylase LSD1 was found to occupy both active and poised enhancers in ES cells⁵⁰. Either the knockdown of LSD1 by RNAi or the use of a covalent inhibitor of LSD1 lead to a retention of H3K4me1 on the *Lefty1* and *Sox2* enhancers and a failure to shut off transcription during differentiation⁵⁰. Since LSD1 co-occupies enhancers with the Mi-2–NuRD nucleosome remodeling and deacetylase complex, it is possible that nucleosome remodeling and histone deacetylation are involved in the decommissioning process⁵⁰.

Although much of the insight obtained into the transitions of enhancer states has come from mapping histone modifications at different developmental stages, it is still not clear to what extent these modifications play a direct role in regulating the function of enhancers. The knockdown or knockout of a factor leading to reduced enhancer function could just be a consequence of the requirement for the enzyme to interact with other transcription factors⁷¹. Indeed, context dependency in the function of Set1/COMPASS and associated factors such as Rad6–Bre1, the Paf1 complex and FACT was recently demonstrated in the yeast *Saccharomyces cerevisiae*⁷². Studies using catalytically inactive enzymes to complement cells depleted of the endogenous enzymes should help clarify this question.

eRNAs and lncRNAs functioning as enhancers

In mammalian cells, most active enhancers exhibit bidirectional transcription^{73,74}. As mentioned earlier, the SV40 enhancer induced transcription

from a promoterless construct, demonstrating the ability of enhancers to recruit Pol II². Transcription at endogenous enhancers was first described for the β -globin and other LCRs (reviewed in ref. 73). Subsequently, genome-wide Pol II ChIP-seq and RNA-seq studies demonstrated that most enhancers show an active chromatin signature and are transcribed by Pol II^{75,76}. These transcripts, named enhancer RNAs (eRNAs), can be polyadenylated, but are not spliced and are generally unstable⁷⁶. Unlike long noncoding RNA genes (lncRNAs), eRNAs do not bear the chromatin signatures of protein-coding genes, such as a prominent mark of H3K4me3 at the start site and H3K36me3 distributed over gene bodies⁷³. Since the appearance of eRNAs precedes the activation of nearby genes, the occurrence of eRNAs has been variously proposed to keep the enhancer region in an open configuration⁷⁷, to begin the process of Pol II tracking from the enhancer to the promoter²³ and/or to facilitate enhancer-promoter looping⁷⁸.

In addition to eRNAs, another type of noncoding RNA contributing to gene activation is a newly discovered class of lncRNA called ncRNA-a that is required to activate the transcription of nearby genes⁷⁹. The ncRNA-a genes can function like enhancers in that they can confer increased expression of a reporter gene from a heterologous promoter⁷⁹. The ncRNA-a transcripts are both spliced and polyadenylated; and the ncRNA-a genes are marked with H3K4me3 and H3K36me3, much like protein-coding genes and the previously described lncRNAs⁷³. RNAi knockdown of ncRNA-a results in reduced enhancer-promoter communication as determined by 3C assays and from reduced expression of nearby genes⁸⁰. Rather than existing solely to keep the enhancer chromatin open, ncRNA-a are proposed to recruit coactivator complexes, as with the recruitment of the MLL complexes by the *HOTTIP* ncRNA to activate *HOX* genes⁸¹. Previously, the vast majority of lncRNA genes were reported

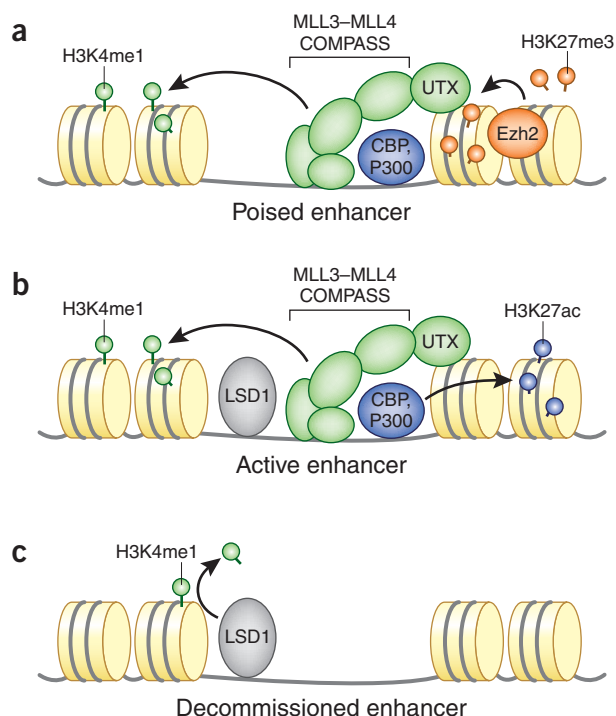


Figure 4 Setting chromatin signatures associated with different enhancer states. (a) Trr in *Drosophila* and MLL3 and MLL4 in mammals have been shown to implement H3K4 monomethylation (H3K4me1) at enhancers, including at enhancers that appear to be in a poised state characterized by the presence of H3K27 trimethylation implemented by E(z) in *Drosophila* and Ezh2 in mammals. (b) Active enhancers, those conferring expression of the associated genes, are characterized by the presence of H3K27ac, instead of H3K27me3, at poised enhancers. Interestingly, CBP and P300, the histone acetyltransferases implementing H3K27ac at active enhancers, are already found at poised enhancers. The transition from H3K27me3 to H3K27ac could be aided by the existence of the H3K27 demethylase UTX in the Trr/MLL3-MLL4 complexes. (c) During differentiation, some enhancers that were active return to a naive state, devoid of the enhancer H3K4me1. This 'decommissioning' process has been attributed to the activity of the H3K4 demethylase LSD1. Only mammalian proteins are shown for simplicity.

these enhancers within 4 hours of LPS treatment. The binding sites for Pu.1 are thought to be weaker at these 'latent' or *de novo* enhancers such that stimulus-dependent transcription factors are needed to cooperate with Pu.1 for accessing its binding sites⁸⁸. A similar study found that Pu.1 and NF- κ B can be detected at *de novo* enhancers as soon as 1 hour after treatment with the lipopolysaccharide TLR4 (ref. 89). Comparison of global nuclear run-on transcription (GRO-Seq) and histone modification ChIP-seq showed that eRNA transcription preceded histone methylation at *de novo* enhancers. Inhibitors of Pol II elongation inhibited the induction of H3K4 methylation, indicating that eRNAs were required for enhancer activation. However, induced degradation of eRNAs with locked nucleic acids had no effect on the induction of H3K4 methylation, suggesting that the transcription process that generates eRNAs is required for the early steps in enhancer activation, but the eRNA products are not⁸⁹.

Enhanceropathies

It has long been known that alteration of enhancer function can lead to disease, with notable examples including the aforementioned deletion of the LCR in thalassemia patients and the translocation of the IgH enhancer in proximity to *MYC* in Burkitt's lymphoma³⁻⁵. However, the malfunctioning of enhancers in human disease not only results from alterations of the enhancer sequences but can also result from a mutation of the proteins that regulate enhancer function. Cornelia de Lange syndrome results from mutations in subunits of the cohesin complex: Smc1, Smc3 or the cohesin-loading factor NIPBL. Although cohesin is best known for its role in chromosome segregation, the developmental defects seen in Cornelia de Lange syndrome are most likely due to cohesin's role in enhancer-promoter communication^{90,91}. Cells can carry on sister chromatid cohesion and chromosome separation when only 13% of the wild-type level of cohesin is present, whereas clinical manifestations of Cornelia de Lange syndrome can occur even with 85% of wild-type amounts of NIPBL⁹². Thus, Cornelia de Lange syndrome is a confirmed cohesinopathy and a candidate enhanceropathy.

Another developmental syndrome involving a protein regulator of enhancer function is Kabuki syndrome⁹³. Kabuki syndrome is characterized by mental retardation and a variety of developmental abnormalities, including the long eyelashes (resembling the prominent eye makeup of Kabuki theater performers) that gave the syndrome its name. Missense or nonsense mutations in *MLL4* (also known as *KMT2D* in humans) were found in seven of ten exomes sequenced⁹³. It is not known to what extent the other cases involve mutations in other genes or in the noncoding regions of *MLL4*. However, the gene encoding H3K27 demethylase UTX, which functions with the Trr/MLL3-MLL4 COMPASS-like complex at enhancers⁵¹, was deleted in 3 of 32 patients with Kabuki syndrome who did not have mutations in *MLL4* (ref. 94).

to recruit PRC2 and other repressive factors for gene silencing, including the *H19* RNA gene involved in imprinting and the *Xist* RNA involved in X-chromosome inactivation⁸².

Pioneer transcription factors prime future gene activation

The initial recognition of enhancers requires pioneer transcription factors⁸³ (Table 1). Pioneer factors, such as GATA, FOXA and Pu.1, can bind their consensus sequence within nucleosomes and facilitate the binding of additional transcription factors to open chromatin in response to hormones or other differentiation signals. For example, the estrogen-dependent DNA binding by the estrogen receptor (ER) requires the pioneer factor FOXA1 (ref. 83). Pioneer factors and other transcription factors occupy enhancers, but not promoters, of lineage-specific genes as early as the ES cell stage^{84,85}. Cohesin, Mediator, P300 and the transcription elongation factor Ell3 have all been shown to occupy enhancers such as the HS2 element of the β -globin LCR in ES cells, even though looping of this enhancer with its promoter does not occur until the erythroid lineage^{86,87}. Therefore, the interaction of DNA-binding transcription factors and other general transcription factors with poised enhancers may be required to set up the poised state. However, the exact mechanism of selectivity of these factors in their target binding sites and enhancer priming is not clear at this time.

For example, we know that the binding of the pioneer factor Foxd3 to the *Alb1* enhancer helps maintain the enhancer in an unmethylated state; RNAi knockdown of Foxd3 increases CpG methylation of the *Alb1* enhancer in ES cells, while forced expression of Foxd3 in mouse embryonic fibroblasts (MEFs) results in the loss of methylation at the *Alb1* enhancer⁸⁵. Although enhancers are not typically associated with CpG islands, methylation of CpG can block transcription-factor binding at enhancers, providing one rationale for priming enhancers early in development.

Enhancer priming by pioneer factors can also occur in differentiated cells. Stimulation of macrophages with lipopolysaccharides (LPS) leads to increased acetylation at a few thousand putative enhancers, but 15% of these enhancers were not previously occupied by Pu.1 or H3K4me1 (ref. 88). Pu.1, H3K4me1 and H3K27ac were all found at

Katie Vicari

In another study of 81 patients with Kabuki syndrome, of the 31 who did not have a mutation in *MLL4*, 5 (16%) had a mutation in *UTX*⁹⁵. Although NIPBL and *MLL4* may have functions outside of enhancers that are responsible for their associated developmental disorders, it is reasonable to consider that a mutation of enhancer regulators could lead to the multiple developmental abnormalities characteristic of these syndromes.

In *Drosophila*, cells with mutations of *Trr*, the homolog of *MLL3* and *MLL4*, or of *Utx* have a growth advantage compared to the surrounding tissue^{96–98}. In humans, *MLL3*, *MLL4* and *UTX* are each frequently mutated in multiple types of cancer^{98–104}. *MLL3* and *MLL4* are known to have some redundancy with each other in the maintenance of global levels of H3K4me1 at enhancers⁵². However, unlike in Kabuki syndrome, where only *MLL4* is known to be mutated, *MLL3* and *MLL4* mutations are found at similar frequencies in cancer⁹⁸. A study of primary colon cancer cell lines found that H3K4me1 was differentially present at the enhancer regions of a signature set of genes when compared to its pattern in normal tissue, suggesting that altered *MLL3* or *MLL4* activity could contribute to oncogenesis¹⁰⁵.

Mutations in *CBP* or *P300*, the products of which acetylate H3K27 at enhancers, can cause Rubinstein-Taybi syndrome, although mutations in *CBP* are about ten times more prevalent⁶³. The fact that the loss of one copy of the gene encoding either enzyme can cause the syndrome suggests that the total protein levels of CBP and P300 are important for development. Similarly, the frequent occurrence of the loss of one copy of *MLL3* or *MLL4* in cancer suggests that their protein levels are very important. CBP and P300 are known to acetylate many proteins, including p53, and to have many cellular roles, including the acetylation at promoters and the acetylation of histone H3 during histone deposition^{106,107}. Therefore, it is not known whether Rubenstein-Taybi syndrome is caused by an alteration of the H3K27ac activity at enhancers.

Mutations in pioneer transcription factors such as members of the GATA family are found in human diseases such as anemia, congenital heart defects and cancer¹⁰⁸. Mutations affecting the activity or levels of GATA2 can lead to the autosomal dominant MonoMAC syndrome, which is characterized by immunodeficiency due to reduced monocytes, B-lymphocytes and natural killer cells. Deletions of the gene or missense mutations within a zinc finger of GATA2 have been identified in patients with MonoMAC¹⁰⁹. Six patients were found to have mutations within an intronic enhancer of *GATA2* causing reduced expression of the affected allele, some consisting of single point mutations within an ETS motif¹¹⁰.

A remarkable example of a point mutation associated with a developmental abnormality is at an enhancer located 1 Mb from the sonic hedgehog gene (*Shh*)¹¹¹. Point mutations within this enhancer can confer ectopic expression of *Shh* and polydactyly in mice, humans and even the Hemingway cats of Key West^{111,112}. The combination of genome-wide association studies (GWAS) and chromatin signatures of enhancers has been used to identify single-nucleotide polymorphisms (SNPs) at enhancers associated with conditions such as cancer, diabetes, rheumatoid arthritis, and Alzheimer's disease^{39,105,113–115}.

The identification of SNPs at enhancers can have implications for the treatment of disease, even when a SNP does not cause disease. This was demonstrated by the recent identification of a single SNP at a location with a chromatin signature associated with a modest increase in fetal hemoglobin¹¹⁶. The widely occurring SNP is located within an intronic enhancer of the *BCL11A* gene encoding a transcriptional repressor of embryonic and fetal hemoglobin expression. Reduced expression of *BCL11A* in the erythroid lineage can ameliorate sickle-cell disease in mouse models¹¹⁷. Transcription activator-like effector nucleases (TALENs) were used to remove a 10-kb region encompassing the enhancer in mouse erythroleukemia cells. *BCL11A* expression was greatly reduced and the ratio of embryonic to adult hemoglobin

was greatly increased¹¹⁶. Thus, a non-disease SNP with a modest, but observable, phenotype can be used to identify lineage-specific enhancers. Genome editing techniques could be used to put these enhancers back into a patient to restore production of a needed protein.

Future directions

During the past three decades, much progress has been made toward understanding the *cis*-regulatory logic of gene expression in metazoan development. Further progress will be greatly aided by some of the technologies developed during the past decade for genomic profiling and genome manipulation. For example, the recent discovery of an oxidized form of 5-methylcytosine (5mC), 5-hydroxymethylcytosine (5hmC), was quickly followed by the development of methods to map its distribution genome wide, whose results indicated that this modification has a key role in preventing the accumulation of the silencing-associated 5mC at enhancers¹¹⁸. Measurement of enhancer-promoter communication has gone from the analysis of a single locus with 3C to the genome-wide mapping of chromosome interactions with next-generation sequencing methods such as Hi-C and ChIA-PET^{46,119}. Testing of individual elements in reporter assays similar to the original SV40 enhancer assay pales in comparison to strategies such as Starr-seq¹²⁰. In this method, the whole genome is sheared and batch tested in reporter assays, with RNA-seq used as a quantitative measure of enhancer strength¹²⁰. Genome editing using TALENs or CRISPR can be used to mutate these newly identified putative enhancers in order to test their function in their natural context¹²¹. Genome editing could be used to modify *cis*-regulatory regions for therapeutic purposes, for example deleting tissue-specific enhancers as has been demonstrated for *BCL11A* in a mouse model¹¹⁶. As an alternative to mutating an enhancer, TAL effectors were used to direct the H3K4 demethylase LSD1 to a particular enhancer for decommissioning¹²².

Even though many factors that regulate enhancers have been identified, not enough is known about how all of these factors work together. For example, multiple factors have been found to function in enhancer-promoter communication, yet it is still not clear how this communication is initially established. It will be important to continue to use cellular stimulation and differentiation systems to determine the order of action and relative necessity of these factors. Although much attention has been paid to enhancer-promoter communication, additional interactions between other *cis*-regulatory elements that fine-tune enhancer function, such as long-range and short-range repressors, need further investigation¹²³. Dissecting some of the old and newly identified extended LCR-like *cis*-regulatory regions and the interaction maps provided by Hi-C and related technologies will provide plenty of examples through which we can further investigate these processes.

Genome-wide mapping of histone modifications has been extremely useful in identifying chromatin signatures at enhancers, and many of the enzymes that implement these modifications have been identified. However, examples exist in which catalytically inactive enzymes are sufficient to provide much of their cellular function^{124–128}. It will be important to determine whether the enzymes and their associated proteins are sufficient or whether modification of histones is also required. Little is known about how and why these enzymes are recruited to enhancers, which often provides the context for their function⁷¹.

The recent identification of widespread transcription at enhancers opens up a new area of investigation toward understanding enhancer function. For example, does enhancer transcription initiate the tracking of Pol II to promoters? Some evidence suggests that the ongoing process of transcription, perhaps the opening of chromatin, and not the RNA molecule is important⁸⁹, whereas other studies suggest that the eRNA product is important¹²⁹. The study of these eRNAs in

more cellular contexts will likely demonstrate diversity of function of eRNA transcription.

There are only a few cases of well-studied lncRNAs functioning in gene regulation, yet genome-wide analyses have identified a large number of lncRNAs for which no function is known. Some of these act like enhancers in that they activate the transcription of nearby genes. Determining the mechanism by which they function is an exciting area of future investigation. Many of these lncRNAs are poorly conserved, suggesting that secondary and tertiary structure is likely to be important. The availability of new genome editing techniques will greatly facilitate investigations into the structure and function of these RNAs *in vivo*.

Answering just some of these questions will lead to a better understanding of the biology of enhancers and the nature of their malfunctioning in disease, which could be invaluable in advancing the treatment of enhanceropathies.

ACKNOWLEDGMENTS

We thank D. Dorsett and H.-M. Herz for helpful discussions, M. Miller for preparation of figures, M. Morgan for the term ‘enhanceropathy’, and L. Shilatfard and L. Kennedy for editorial assistance. The work on enhancers in the Shilatfard laboratory is funded by US National Institutes of Health grants R01CA150265 and R01CA89455.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

Reprints and permissions information is available online at <http://www.nature.com/reprints/index.html>.

1. Banerji, J., Rusconi, S. & Schaffner, W. Expression of a beta-globin gene is enhanced by remote SV40 DNA sequences. *Cell* **27**, 299–308 (1981).
2. Moreau, P. *et al.* The SV40 72 base repair repeat has a striking effect on gene expression both in SV40 and other chimeric recombinants. *Nucleic Acids Res.* **9**, 6047–6068 (1981).
- Refs. 1 and 2 reported the identification of the SV40 enhancer whose properties became defining features of enhancers for the next 30 years.**
3. Banerji, J., Olson, L. & Schaffner, W. A lymphocyte-specific cellular enhancer is located downstream of the joining region in immunoglobulin heavy chain genes. *Cell* **33**, 729–740 (1983).
4. Gillies, S.D., Morrison, S.L., Oi, V.T. & Tonegawa, S. A tissue-specific transcription enhancer element is located in the major intron of a rearranged immunoglobulin heavy chain gene. *Cell* **33**, 717–728 (1983).
5. Mercola, M., Wang, X.F., Olsen, J. & Calame, K. Transcriptional enhancer elements in the mouse immunoglobulin heavy chain locus. *Science* **221**, 663–665 (1983).
- Refs. 3–5 reported the first cellular enhancer, which, when placed in proximity to the MYC gene, leads to Burkitt’s lymphoma.**
6. Taub, R. *et al.* Translocation of the c-myc gene into the immunoglobulin heavy chain locus in human Burkitt lymphoma and murine plasmacytoma cells. *Proc. Natl. Acad. Sci. USA* **79**, 7837–7841 (1982).
7. Schroeder, M.D., Greer, C. & Gaul, U. How to make stripes: deciphering the transition from non-periodic to periodic patterns in Drosophila segmentation. *Development* **138**, 3067–3078 (2011).
8. Levine, M. Transcriptional enhancers in animal development and evolution. *Curr. Biol.* **20**, R754–R763 (2010).
9. Klingler, M., Soong, J., Butler, B. & Gergen, J.P. Disperse versus compact elements for the regulation of runt stripes in Drosophila. *Dev. Biol.* **177**, 73–84 (1996).
10. Jack, J., Dorsett, D., Delotto, Y. & Liu, S. Expression of the cut locus in the Drosophila wing margin is required for cell type specification and is regulated by a distant enhancer. *Development* **113**, 735–747 (1991).
11. Spana, C., Harrison, D.A. & Corces, V.G. The Drosophila melanogaster suppressor of Hairy-wing protein binds to specific sequences of the gypsy retrotransposon. *Genes Dev.* **2**, 1414–1423 (1988).
- This study showed that Su(Hw) directly bound sequences in the gypsy insulator, thus providing a mechanistic understanding for years of genetic studies and helping to launch the new field of insulator biology.**
12. Morcillo, P., Rosen, C., Baylies, M.K. & Dorsett, D. Chip, a widely expressed chromosomal protein required for segmentation and activity of a remote wing margin enhancer in Drosophila. *Genes Dev.* **11**, 2729–2740 (1997).
13. Morcillo, P., Rosen, C. & Dorsett, D. Genes regulating the remote wing margin enhancer in the Drosophila cut locus. *Genetics* **144**, 1143–1154 (1996).
- This study used a genetic screen for factors regulating enhancer-promoter communication and identified genes later shown to encode the fly homologs of NIPBL and LDB1. These proteins were subsequently shown to be involved in the formation and/or maintenance of enhancer-promoter looping in mammals.**
14. Rollins, R.A., Korom, M., Aulner, N., Martens, A. & Dorsett, D. Drosophila nipped-B protein supports sister chromatid cohesion and opposes the stromalin/Scc3 cohesion factor to facilitate long-range activation of the cut gene. *Mol. Cell. Biol.* **24**, 3100–3111 (2004).

15. Rollins, R.A., Morcillo, P. & Dorsett, D. Nipped-B, a Drosophila homologue of chromosomal adherens, participates in activation by remote enhancers in the cut and Ultrabithorax genes. *Genetics* **152**, 577–593 (1999).
16. Agulnick, A.D. *et al.* Interactions of the LIM-domain-binding factor Ldb1 with LIM homeodomain proteins. *Nature* **384**, 270–272 (1996).
17. Lee, S.K. & Pfaff, S.L. Synchronization of neurogenesis and motor neuron specification by direct coupling of bHLH and homeodomain transcription factors. *Neuron* **38**, 731–745 (2003).
18. Deng, W. *et al.* Controlling long-range genomic interactions at a native locus by targeted tethering of a looping factor. *Cell* **149**, 1233–1244 (2012).
- This study provided direct experimental confirmation that LDB1 bridges enhancer-promoter communication.**
19. Torigoi, E. *et al.* Chip interacts with diverse homeodomain proteins and potentiates bicoid activity in vivo. *Proc. Natl. Acad. Sci. USA* **97**, 2686–2691 (2000).
20. Soler, E. *et al.* The genome-wide dynamics of the binding of Ldb1 complexes during erythroid differentiation. *Genes Dev.* **24**, 277–289 (2010).
21. Kagey, M.H. *et al.* Mediator and cohesin connect gene expression and chromatin architecture. *Nature* **467**, 430–435 (2010).
- This study used genome-wide analyses and chromatin conformation assays to reveal the general requirement of cohesin in enhancer-promoter communication in mouse embryonic stem cells.**
22. Schaaf, C.A. *et al.* Genome-wide control of RNA polymerase II activity by cohesin. *PLoS Genet.* **9**, e1003382 (2013).
23. Bulger, M. & Groudine, M. Functional and mechanistic diversity of distal transcription enhancers. *Cell* **144**, 327–339 (2011).
24. Kulaeva, O.I., Nizovtseva, E.V., Polikanov, Y.S., Ulianov, S.V. & Studitsky, V.M. Distant activation of transcription: mechanisms of enhancer action. *Mol. Cell. Biol.* **32**, 4892–4897 (2012).
25. Williams, T.N. & Weatherall, D.J. World distribution, population genetics, and health burden of the hemoglobinopathies. *Cold Spring Harb. Perspect. Med.* **2**, a011692 (2012).
26. Kioussis, D., Vanin, E., deLange, T., Flavell, R.A. & Grosfeld, F.G. β -globin gene inactivation by DNA translocation in $\gamma\beta$ -thalassaemia. *Nature* **306**, 662–666 (1983).
27. Van der Ploeg, L.H. *et al.* $\gamma\beta$ -Thalassaemia studies showing that deletion of the γ - and δ -genes influences β -globin gene expression in man. *Nature* **283**, 637–642 (1980).
- Refs. 26 and 27 helped link a deletion in the DNase-hypersensitive β -globin locus control region with thalassemias.**
28. Kioussis, D. & Festenstein, R. Locus control regions: overcoming heterochromatin-induced gene inactivation in mammals. *Curr. Opin. Genet. Dev.* **7**, 614–619 (1997).
29. Fraser, P. & Grosfeld, F. Locus control regions, chromatin activation and transcription. *Curr. Opin. Cell Biol.* **10**, 361–365 (1998).
30. Gilbert, S.F. *Developmental Biology* (Sinauer Associates, Sunderland, Massachusetts, USA, 2000).
31. Bonifer, C., Vidal, M., Grosfeld, F. & Sippel, A.E. Tissue specific and position independent expression of the complete gene domain for chicken lysozyme in transgenic mice. *EMBO J.* **9**, 2843–2848 (1990).
32. Jones, B.K., Monks, B.R., Liebhaber, S.A. & Cooke, N.E. The human growth hormone gene is regulated by a multicomponent locus control region. *Mol. Cell. Biol.* **15**, 7010–7021 (1995).
33. modENCODE Consortium *et al.* Identification of functional elements and regulatory circuits by Drosophila modENCODE. *Science* **330**, 1787–1797 (2010).
34. Kvon, E.Z., Stampfel, G., Yanez-Cuna, J.O., Dickson, B.J. & Stark, A. HOT regions function as patterned developmental enhancers and have a distinct cis-regulatory signature. *Genes Dev.* **26**, 908–913 (2012).
35. Lovén, J. *et al.* Selective inhibition of tumor oncogenes by disruption of super-enhancers. *Cell* **153**, 320–334 (2013).
36. Whyte, W.A. *et al.* Master transcription factors and mediator establish super-enhancers at key cell identity genes. *Cell* **153**, 307–319 (2013).
37. Koch, F. *et al.* Transcription initiation platforms and GTF recruitment at tissue-specific enhancers and promoters. *Nat. Struct. Mol. Biol.* **18**, 956–963 (2011).
38. Parker, S.C. *et al.* Chromatin stretch enhancer states drive cell-specific gene regulation and harbor human disease risk variants. *Proc. Natl. Acad. Sci. USA* **110**, 17921–17926 (2013).
39. Hnisz, D. *et al.* Super-enhancers in the control of cell identity and disease. *Cell* **155**, 934–947 (2013).
40. Barolo, S. Shadow enhancers: frequently asked questions about distributed cis-regulatory information and enhancer redundancy. *BioEssays* **34**, 135–141 (2012).
41. Lagha, M., Bothma, J.P. & Levine, M. Mechanisms of transcriptional precision in animal development. *Trends Genet.* **28**, 409–416 (2012).
42. Perry, M.W., Boettiger, A.N., Bothma, J.P. & Levine, M. Shadow enhancers foster robustness of Drosophila gastrulation. *Curr. Biol.* **20**, 1562–1567 (2010).
43. Hong, J.W., Hendrix, D.A. & Levine, M.S. Shadow enhancers as a source of evolutionary novelty. *Science* **321**, 1314 (2008).
- This study used genome-wide occupancy of transcription factors in Drosophila to identify novel, seemingly redundant enhancers for well-known enhancers of developmental genes.**
44. Frankel, N. *et al.* Phenotypic robustness conferred by apparently redundant transcriptional enhancers. *Nature* **466**, 490–493 (2010).
45. Casellas, R., Yamane, A., Kovalchuk, A.L. & Potter, M. Restricting activation-induced cytidine deaminase tumorigenic activity in B lymphocytes. *Immunology* **126**, 316–328 (2009).
46. Kieffer-Kwon, K.R. *et al.* Interactome maps of mouse gene regulatory domains reveal basic principles of transcriptional regulation. *Cell* **155**, 1507–1520 (2013).
- This study found that deletion of either the E1 or E2 enhancers for the *Aicd* gene led to the abolishment of *Aicd* expression. The authors suggested that genes whose induction needs to be tightly controlled require two enhancers to be turned on, a system referred to in this Review as fail-safe or split enhancers.**

47. Heintzman, N.D. *et al.* Distinct and predictive chromatin signatures of transcriptional promoters and enhancers in the human genome. *Nat. Genet.* **39**, 311–318 (2007).
48. Creighton, M.P. *et al.* Histone H3K27ac separates active from poised enhancers and predicts developmental state. *Proc. Natl. Acad. Sci. USA* **107**, 21931–21936 (2010).
49. Rada-Iglesias, A. *et al.* A unique chromatin signature uncovers early developmental enhancers in humans. *Nature* **470**, 279–283 (2011).
Refs. 47–49 used genome-wide mapping of histone modifications to identify chromatin signatures for different states of enhancer activity.
50. Whyte, W.A. *et al.* Enhancer decommisioning by LSD1 during embryonic stem cell differentiation. *Nature* **482**, 221–225 (2012).
51. Herz, H.M. *et al.* Enhancer-associated H3K4 monomethylation by Trithorax-related, the Drosophila homolog of mammalian Mll3/Mll4. *Genes Dev.* **26**, 2604–2620 (2012).
52. Hu, D. *et al.* The MLL3/MLL4 branches of the COMPASS family function as major histone H3K4 monomethylases at enhancers. *Mol. Cell. Biol.* **33**, 4745–4754 (2013).
Refs. 51 and 52 identified Drosophila Trl and its mammalian homologs MLL3 and MLL4 as major H3K4 monomethylases at enhancers.
53. Shilatifard, A. The COMPASS family of histone H3K4 methylases: mechanisms of regulation in development and disease pathogenesis. *Annu. Rev. Biochem.* **81**, 65–95 (2012).
54. Ardehali, M.B. *et al.* Drosophila Set1 is the major histone H3 lysine 4 trimethyltransferase with role in transcription. *EMBO J.* **30**, 2817–2828 (2011).
55. Mohan, M. *et al.* The COMPASS family of H3K4 methylases in Drosophila. *Mol. Cell. Biol.* **31**, 4310–4318 (2011).
56. Eissenberg, J.C. & Shilatifard, A. Histone H3 lysine 4 (H3K4) methylation in development and differentiation. *Dev. Biol.* **339**, 240–249 (2010).
57. Sedkov, Y. *et al.* Molecular genetic analysis of the Drosophila trithorax-related gene which encodes a novel SET domain protein. *Mech. Dev.* **82**, 171–179 (1999).
58. Wang, P. *et al.* Global analysis of H3K4 methylation defines MLL family member targets and points to a role for MLL1-mediated H3K4 methylation in the regulation of transcriptional initiation by RNA polymerase II. *Mol. Cell. Biol.* **29**, 6074–6085 (2009).
59. Yu, B.D., Hess, J.L., Horning, S.E., Brown, G.A. & Korsmeyer, S.J. Altered Hox expression and segmental identity in Mll-mutant mice. *Nature* **378**, 505–508 (1995).
60. Tie, F. *et al.* CBP-mediated acetylation of histone H3 lysine 27 antagonizes Drosophila Polycomb silencing. *Development* **136**, 3131–3141 (2009).
61. Jin, Q. *et al.* Distinct roles of GCN5/PCAF-mediated H3K9ac and CBP/p300-mediated H3K18/27ac in nuclear receptor transactivation. *EMBO J.* **30**, 249–262 (2011).
62. Wang, L., Tang, Y., Cole, P.A. & Marmorstein, R. Structure and chemistry of the p300/CBP and Rtt109 histone acetyltransferases: implications for histone acetyltransferase evolution and function. *Curr. Opin. Struct. Biol.* **18**, 741–747 (2008).
63. Roelfsema, J.H. *et al.* Genetic heterogeneity in Rubinstein-Taybi syndrome: mutations in both the CBP and EP300 genes cause disease. *Am. J. Hum. Genet.* **76**, 572–580 (2005).
64. Roth, J.F. *et al.* Differential role of p300 and CBP acetyltransferase during myogenesis: p300 acts upstream of MyoD and Myf5. *EMBO J.* **22**, 5186–5196 (2003).
65. Kharchenko, P.V. *et al.* Comprehensive analysis of the chromatin landscape in *Drosophila melanogaster*. *Nature* **471**, 480–485 (2011).
66. Di Croce, L. & Helin, K. Transcriptional regulation by Polycomb group proteins. *Nat. Struct. Mol. Biol.* **20**, 1147–1155 (2013).
67. Schwartz, Y.B. & Pirrotta, V. Polycomb silencing mechanisms and the management of genomic programmes. *Nat. Rev. Genet.* **8**, 9–22 (2007).
68. Herz, H.M., Garruss, A. & Shilatifard, A. SET for life: biochemical activities and biological functions of SET domain-containing proteins. *Trends Biochem. Sci.* **38**, 621–639 (2013).
69. Heintzman, N.D. *et al.* Histone modifications at human enhancers reflect global cell-type-specific gene expression. *Nature* **459**, 108–112 (2009).
70. Tie, F., Banerjee, R., Conrad, P.A., Scacheri, P.C. & Harte, P.J. Histone demethylase UTX and chromatin remodeler BRM bind directly to CBP and modulate acetylation of histone H3 lysine 27. *Mol. Cell. Biol.* **32**, 2323–2334 (2012).
71. Lee, J.S., Smith, E. & Shilatifard, A. The language of histone crosstalk. *Cell* **142**, 682–685 (2010).
72. Thornton, J.L. *et al.* Context dependency of Set1/COMPASS-mediated histone H3 Lys4 trimethylation. *Genes Dev.* **28**, 115–120 (2014).
73. Natoli, G. & Andrau, J.C. Noncoding transcription at enhancers: general principles and functional models. *Annu. Rev. Genet.* **46**, 1–19 (2012).
74. Ørom, U.A. & Shiekhattar, R. Long noncoding RNAs usher in a new era in the biology of enhancers. *Cell* **154**, 1190–1193 (2013).
75. Kim, T.K. *et al.* Widespread transcription at neuronal activity-regulated enhancers. *Nature* **465**, 182–187 (2010).
76. De Santa, F. *et al.* A large fraction of extragenic RNA pol II transcription sites overlap enhancers. *PLoS Biol.* **8**, e1000384 (2010).
77. Ashe, H.L., Monks, J., Wijgerde, M., Fraser, P. & Proudfoot, N.J. Intergenic transcription and transinduction of the human β -globin locus. *Genes Dev.* **11**, 2494–2509 (1997).
78. Li, W. *et al.* Functional roles of enhancer RNAs for oestrogen-dependent transcriptional activation. *Nature* **498**, 516–520 (2013).
79. Ørom, U.A. *et al.* Long noncoding RNAs with enhancer-like function in human cells. *Cell* **143**, 46–58 (2010).
Refs. 78 and 79 showed that some of the most active enhancers are transcribed.
80. Lai, F. *et al.* Activating RNAs associate with Mediator to enhance chromatin architecture and transcription. *Nature* **494**, 497–501 (2013).
81. Wang, K.C. *et al.* A long noncoding RNA maintains active chromatin to coordinate homeotic gene expression. *Nature* **472**, 120–124 (2011).
Refs. 79–81 found that polyadenylated lncRNAs could activate promoters of nearby genes. When placed in reporter constructs, these RNAs have enhancer-like function, revealing a previously unknown diversity in the cis-regulatory landscape.
82. Rinn, J.L. & Chang, H.Y. Genome regulation by long noncoding RNAs. *Annu. Rev. Biochem.* **81**, 145–166 (2012).
83. Zaret, K.S. & Carroll, J.S. Pioneer transcription factors: establishing competence for gene expression. *Genes Dev.* **25**, 2227–2241 (2011).
84. Liber, D. *et al.* Epigenetic priming of a pre-B cell-specific enhancer through binding of Sox2 and Foxd3 at the ESC stage. *Cell Stem Cell* **7**, 114–126 (2010).
85. Xu, J. *et al.* Transcriptional competence and the active marking of tissue-specific enhancers by defined transcription factors in embryonic and induced pluripotent stem cells. *Genes Dev.* **23**, 2824–2838 (2009).
86. Lin, C., Garruss, A.S., Luo, Z., Guo, F. & Shilatifard, A. The RNA Pol II elongation factor E13 marks enhancers in ES cells and primes future gene activation. *Cell* **152**, 144–156 (2013).
87. Hou, C., Dale, R. & Dean, A. Cell type specificity of chromatin organization mediated by CTCF and cohesin. *Proc. Natl. Acad. Sci. USA* **107**, 3651–3656 (2010).
88. Ostuni, R. *et al.* Latent enhancers activated by stimulation in differentiated cells. *Cell* **152**, 157–171 (2013).
89. Kaikkonen, M.U. *et al.* Remodeling of the enhancer landscape during macrophage activation is coupled to enhancer transcription. *Mol. Cell* **51**, 310–325 (2013).
This study showed that the ongoing process of transcription at an enhancer, but not the eRNA product, was required for enhancer function.
90. Liu, J. & Krantz, I.D. Cohesin and human disease. *Annu. Rev. Genomics Hum. Genet.* **9**, 303–320 (2008).
91. Dorsett, D. & Krantz, I.D. On the molecular etiology of Cornelia de Lange syndrome. *Ann. NY Acad. Sci.* **1151**, 22–37 (2009).
92. Borck, G. *et al.* Father-to-daughter transmission of Cornelia de Lange syndrome caused by a mutation in the 5' untranslated region of the NIPBL gene. *Hum. Mutat.* **27**, 731–735 (2006).
93. Ng, S.B. *et al.* Exome sequencing identifies MLL2 mutations as a cause of Kabuki syndrome. *Nat. Genet.* **42**, 790–793 (2010).
94. Lederer, D. *et al.* Deletion of KDM6A, a histone demethylase interacting with MLL2, in three patients with Kabuki syndrome. *Am. J. Hum. Genet.* **90**, 119–124 (2012).
95. Miyake, N. *et al.* MLL2 and KDM6A mutations in patients with Kabuki syndrome. *Am. J. Med. Genet. A* **161**, 2234–2243 (2013).
96. Herz, H.M. *et al.* The H3K27me3 demethylase dUTX is a suppressor of Notch- and Rb-dependent tumors in Drosophila. *Mol. Cell. Biol.* **30**, 2485–2497 (2010).
97. Kanda, H., Nguyen, A., Chen, L., Okano, H. & Hariharan, I.K. The Drosophila ortholog of MLL3 and MLL4, trithorax related, functions as a negative regulator of tissue growth. *Mol. Cell. Biol.* **33**, 1702–1710 (2013).
98. Morgan, M.A. & Shilatifard, A. Drosophila SETs its sights on cancer: Trl/MLL3/4 COMPASS-like complexes in development and disease. *Mol. Cell. Biol.* **33**, 1698–1701 (2013).
99. Grasso, C.S. *et al.* The mutational landscape of lethal castration-resistant prostate cancer. *Nature* **487**, 239–243 (2012).
100. Jones, D.T. *et al.* Dissecting the genomic complexity underlying medulloblastoma. *Nature* **488**, 100–105 (2012).
101. Morin, R.D. *et al.* Frequent mutation of histone-modifying genes in non-Hodgkin lymphoma. *Nature* **476**, 298–303 (2011).
102. Parsons, D.W. *et al.* The genetic landscape of the childhood cancer medulloblastoma. *Science* **331**, 435–439 (2011).
103. Pasqualucci, L. *et al.* Analysis of the coding genome of diffuse large B-cell lymphoma. *Nat. Genet.* **43**, 830–837 (2011).
104. Pugh, T.J. *et al.* Medulloblastoma exome sequencing uncovers subtype-specific somatic mutations. *Nature* **488**, 106–110 (2012).
105. Akhtar-Zaidi, B. *et al.* Epigenomic enhancer profiling defines a signature of colon cancer. *Science* **336**, 736–739 (2012).
106. Das, C., Lucia, M.S., Hansen, K.C. & Tyler, J.K. CBP/p300-mediated acetylation of histone H3 on lysine 56. *Nature* **459**, 113–117 (2009).
107. Wang, F., Marshall, C.B. & Ikura, M. Transcriptional/epigenetic regulator CBP/p300 in tumorigenesis: structural and functional versatility in target recognition. *Cell. Mol. Life Sci.* **70**, 3989–4008 (2013).
108. Zheng, R. & Blobel, G.A. GATA transcription factors and cancer. *Genes Cancer* **1**, 1178–1188 (2010).
109. Hsu, A.P. *et al.* Mutations in GATA2 are associated with the autosomal dominant and sporadic monocytopenia and mycobacterial infection (MonoMAC) syndrome. *Blood* **118**, 2653–2655 (2011).
110. Hsu, A.P. *et al.* GATA2 haploinsufficiency caused by mutations in a conserved intronic element leads to MonoMAC syndrome. *Blood* **121**, 3830–3837 (2013).
111. Lettice, L.A. *et al.* A long-range Shh enhancer regulates expression in the developing limb and fin and is associated with preaxial polydactyly. *Hum. Mol. Genet.* **12**, 1725–1735 (2003).
112. Lettice, L.A., Hill, A.E., Devenney, P.S. & Hill, R.E. Point mutations in a distant sonic hedgehog cis-regulator generate a variable regulatory output responsible for preaxial polydactyly. *Hum. Mol. Genet.* **17**, 978–985 (2008).
113. Corradin, O. *et al.* Combinatorial effects of multiple enhancer variants in linkage disequilibrium dictate levels of gene expression to confer susceptibility to common traits. *Genome Res.* **24**, 1–13 (2013).
114. Trynka, G. *et al.* Chromatin marks identify critical cell types for fine mapping complex trait variants. *Nat. Genet.* **45**, 124–130 (2013).

115. Encode Project Consortium *et al.* An integrated encyclopedia of DNA elements in the human genome. *Nature* **489**, 57–74 (2012).
116. Bauer, D.E. *et al.* An erythroid enhancer of BCL11A subject to genetic variation determines fetal hemoglobin level. *Science* **342**, 253–257 (2013).
117. Xu, J. *et al.* Correction of sickle cell disease in adult mice by interference with fetal hemoglobin silencing. *Science* **334**, 993–996 (2011).
These authors deleted a tissue-specific enhancer of an essential gene encoding a transcriptional repressor for fetal hemoglobin and found that this disruption restored globin expression in a mouse model of sickle-cell anemia.
118. Pastor, W.A., Aravind, L. & Rao, A. TETonic shift: biological roles of TET proteins in DNA demethylation and transcription. *Nat. Rev. Mol. Cell Biol.* **14**, 341–356 (2013).
119. Jin, F. *et al.* A high-resolution map of the three-dimensional chromatin interactome in human cells. *Nature* **503**, 290–294 (2013).
120. Arnold, C.D. *et al.* Genome-wide quantitative enhancer activity maps identified by STARR-seq. *Science* **339**, 1074–1077 (2013).
This study developed a genome-wide screen to identify regions of DNA that function in enhancer assays in *Drosophila* S2 cells.
121. Gaj, T., Gersbach, C.A. & Barbas, C.F. III. ZFN, TALEN, and CRISPR/Cas-based methods for genome engineering. *Trends Biotechnol.* **31**, 397–405 (2013).
122. Mendenhall, E.M. *et al.* Locus-specific editing of histone modifications at endogenous enhancers. *Nat. Biotechnol.* **31**, 1133–1136 (2013).
This study used TAL effectors to target LSD1 to silence an enhancer. This approach can help determine the *in vivo* function of an enhancer and could potentially be used for therapeutic purposes.
123. Li, L.M. & Arnosti, D.N. Long- and short-range transcriptional repressors induce distinct chromatin states on repressed genes. *Curr. Biol.* **21**, 406–412 (2011).
124. Li, L., Greer, C., Eisenman, R.N. & Seemba, J. Essential functions of the histone demethylase lid. *PLoS Genet.* **6**, e1001221 (2010).
125. Morales Torres, C., Laugesen, A. & Helin, K. Utx is required for proper induction of ectoderm and mesoderm during differentiation of embryonic stem cells. *PLoS ONE* **8**, e60020 (2013).
126. Shpargel, K.B., Sengoku, T., Yokoyama, S. & Magnuson, T. UTX and UTY demonstrate histone demethylase-independent function in mouse embryonic development. *PLoS Genet.* **8**, e1002964 (2012).
127. Terranova, R., Agherbi, H., Boned, A., Meresse, S. & Djabali, M. Histone and DNA methylation defects at Hox genes in mice expressing a SET domain-truncated form of Mll. *Proc. Natl. Acad. Sci. USA* **103**, 6629–6634 (2006).
128. Wang, C. *et al.* UTX regulates mesoderm differentiation of embryonic stem cells independent of H3K27 demethylase activity. *Proc. Natl. Acad. Sci. USA* **109**, 15324–15329 (2012).
129. Lam, M.T.Y. *et al.* Rev-Erbs repress macrophage gene expression by inhibiting enhancer-directed transcription. *Nature* **498**, 511–515 (2013).
This study demonstrated that an eRNA product has regulatory function.
130. Le Borgne, R., Remaud, S., Hamel, S. & Schweisguth, F. Two distinct E3 ubiquitin ligases have complementary functions in the regulation of delta and serrate signaling in *Drosophila*. *PLoS Biol.* **3**, e96 (2005).
131. Chlon, T.M. & Crispino, J.D. Combinatorial regulation of tissue specification by GATA and FOG factors. *Development* **139**, 3905–3916 (2012).
132. Cirillo, L.A. *et al.* Binding of the winged-helix transcription factor HNF3 to a linker histone site on the nucleosome. *EMBO J.* **17**, 244–254 (1998).
This study demonstrated that the pioneer factor HNF3 not only has a structure resembling that of linker histone H1, but binds to nucleosomal DNA in a similar manner.
133. Cuesta, I., Zaret, K.S. & Santisteban, P. The forkhead factor FoxE1 binds to the thyroperoxidase promoter during thyroid cell differentiation and modifies compacted chromatin structure. *Mol. Cell Biol.* **27**, 7302–7314 (2007).
134. Hatta, M. & Cirillo, L.A. Chromatin opening and stable perturbation of core histone:DNA contacts by FoxO1. *J. Biol. Chem.* **282**, 35583–35593 (2007).
135. Mandel, E.M. & Grosschedl, R. Transcription control of early B cell differentiation. *Curr. Opin. Immunol.* **22**, 161–167 (2010).
136. Natoli, G., Ghisletti, S. & Barozzi, I. The genomic landscapes of inflammation. *Genes Dev.* **25**, 101–106 (2011).
137. Harrison, M.M., Li, X.Y., Kaplan, T., Botchan, M.R. & Eisen, M.B. Zelda binding in the early *Drosophila melanogaster* embryo marks regions subsequently activated at the maternal-to-zygotic transition. *PLoS Genet.* **7**, e1002266 (2011).
138. Liang, H.L. *et al.* The zinc-finger protein Zelda is a key activator of the early zygotic genome in *Drosophila*. *Nature* **456**, 400–403 (2008).
139. Nien, C.Y. *et al.* Temporal coordination of gene networks by Zelda in the early *Drosophila* embryo. *PLoS Genet.* **7**, e1002339 (2011).
140. Dorsett, D. & Merkenschlager, M. Cohesin at active genes: a unifying theme for cohesin and gene expression from model organisms to humans. *Curr. Opin. Cell Biol.* **25**, 327–333 (2013).
141. Matthews, J.M. & Visvader, J.E. LIM-domain-binding protein 1: a multifunctional cofactor that interacts with diverse proteins. *EMBO Rep.* **4**, 1132–1137 (2003).
142. Ghirlando, R. *et al.* Chromatin domains, insulators, and the regulation of gene expression. *Biochim. Biophys. Acta* **1819**, 644–651 (2012).
143. Chetverina, D., Aoki, T., Erokhin, M., Georgiev, P. & Schedl, P. Making connections: insulators organize eukaryotic chromosomes into independent cis-regulatory networks. *BioEssays* **36**, 163–172 (2014).
144. Phillips-Cremins, J.E. & Corces, V.G. Chromatin insulators: linking genome organization to cellular function. *Mol. Cell* **50**, 461–474 (2013).
145. Guo, C. *et al.* KMT2D maintains neoplastic cell proliferation and global histone H3 lysine 4 monomethylation. *Oncotarget* **4**, 2144–2153 (2013).
146. Calo, E. & Wysocka, J. Modification of enhancer chromatin: what, how, and why? *Mol. Cell* **49**, 825–837 (2013).
147. Krebs, A.R., Karmodiya, K., Lindahl-Allen, M., Struhl, K. & Tora, L. SAGA and ATAC histone acetyl transferase complexes regulate distinct sets of genes and ATAC defines a class of p300-independent enhancers. *Mol. Cell* **44**, 410–423 (2011).
148. Zippo, A. *et al.* Histone crosstalk between H3S10ph and H4K16ac generates a histone code that mediates transcription elongation. *Cell* **138**, 1122–1136 (2009).
149. Shi, J. *et al.* Role of SWI/SNF in acute leukemia maintenance and enhancer-mediated Myc regulation. *Genes Dev.* **27**, 2648–2662 (2013).
150. Belkina, A.C. & Denis, G.V. BET domain co-regulators in obesity, inflammation and cancer. *Nat. Rev. Cancer* **12**, 465–477 (2012).